

BY 529: Mechanobiology of the Cell (MBoC)

Course Code	BY 529
Course Name	Mechanobiology of the Cell (MBoC)
L-T-P-C	3-0-0-3
Prerequisites	BE201 Cell Biology or Consent of Faculty Member
Intended for	B.Tech.-M.Tech. Integrated Dual Degree in Bio-Engineering, M. Tech Biotechnology and PhD candidates
Mutual Exclusion	NA
Approval	23rd BoA (Apr 2018)

Course Contents

- **Introduction to Mechanobiology:** Why study mechanobiology, pioneering experiments in mechanobiology. (2 Lectures)
- **Molecular Mechanisms of Mechanotransduction:** Mechanosensory molecules in focal adhesions, cell-cell junctions, cytoskeleton, and nucleus. (6 Lectures)
- **Mechanobiology of cell behavior:** Rigidity sensing and mechanotransduction in adhesion, migration, gene expression, and tissue development. (8 Lectures)
- **Mechanobiology of organ systems:** Cardiovascular, Bone, Cartilage, Liver, Nervous system. (8 Lectures)
- **Mechanobiology of disease:** Muscular dystrophy, cancer, laminopathy. (6 Lectures)
- **Technology innovation for mechanobiology:** Optical microscopy, nanofabrication, microfluidics, organoids, organ-on-chip. (6 Lectures)
- **Mechanobiology in medical diagnostics and therapeutics:** Cell therapy, Cancer diagnostics, Immune profiling. (6 Lectures)

Text Books

1. Christopher R. Jacobs, Hayden Huang, and Ronal Y. Kwon, **Introduction to Cell Mechanics and Mechanobiology**, Taylor & Francis Group, 2012.
2. Michael Sheetz and Harry Yu, **The Cell as a Machine**, Cambridge University Press, 2018

References

1. Ronen Zaidel-Bar, **Mechanobiology: Methods and Protocols**, Humana Press, Springer Protocols, Methods in Molecular Biology 2023
2. Stefaan W. Verbruggen, **Mechanobiology in Health and Disease**, Academic Press, 2018.
3. MBIInfo (<https://www.mbi.nus.edu.sg/mbinfo/>) This is a wiki-style repository of mechanobiology with the aim to inform and educate the wider scientific community about mechanobiology and how physics and mechanics impact biological processes.